

Pavan Nagesh

9663357603 | pavannagesh681@gmail.com | linkedin.com/in/pavan-nagesh | github.com/PavanNagesh | Hubli, Karnataka

Software Engineer | Full-Stack & AI Engineering

SUMMARY

Computer Science Engineering student with hands-on experience in full-stack development, machine learning, and generative AI across freelance, internship, and hackathon settings. Proficient in Python, JavaScript, React, and Node.js, with applied experience building REST APIs, microservices, and RAG-based LLM pipelines. Delivered measurable impact, including an 80% reduction in manual workflow time and a national Top-3 hackathon finish. Seeking a Software Engineer role to build scalable, production-grade systems.

EDUCATION

KLE Technological University

Bachelor of Engineering in Computer Science

Hubli, Karnataka

2024 – 2027

KLE Smt C.I.M Polytechnic College

Diploma in Computer Science

Hubli, Karnataka

2021 – 2024

SKILLS

Languages: Python, C++, JavaScript, HTML, CSS

Frameworks & Libraries: React.js, Node.js, FastAPI, LangChain, Flutter

AI / Machine Learning: Machine Learning (ML), Generative AI (GenAI), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Exploratory Data Analysis (EDA)

Backend & Architecture: REST APIs, Microservices Architecture, Full-Stack Development, Blockchain

Databases: MongoDB, MSSQL, MySQL

Tools & Platforms: Git, Linux, Ansible, CI/CD, Vercel

EXPERIENCE

Freelance Full-Stack Developer – Klean Tech Systems

May 2026

Built and shipped SmartQuote, a production quotation and invoice automation platform for a retail business

Hubli, India

- Developed and deployed a SaaS platform used to manage 1,000+ products and generate quotations, invoices, and customer records for daily business operations.
- Implemented Excel-based bulk imports, automated GST calculations, and CI/CD on Vercel, reducing quotation preparation time by 80% and minimizing manual errors.

AI Research Intern – Traction Layer AI

Sep 2025 – Mar 2026

Built an Ambient AI platform for real-time clinical documentation and physician workflow automation

Hubli, India

- Developed an end-to-end AI pipeline with multilingual speech-to-text, speaker diarization, and LLM-based SOAP note generation to automate clinical documentation.
- Implemented a multi-agent RAG architecture to generate context-aware clinical notes grounded in hospital knowledge bases while ensuring accuracy and privacy compliance.

Project Intern – Infosys

Aug 2025 – Jan 2026

Built a decentralized slot booking system using Beckn Protocol

Hubli, India

- Architected a Beckn-compliant microservices system (Gateway, BAP, BPP, and AI Service) for decentralized slot booking.
- Designed asynchronous request-response workflows and orchestrated inter-service communication following Beckn protocol standards.

PROJECTS

AI/ML Consensus-Aware Blockchain Simulator | Blockchain, Machine Learning | Python, Scikit-learn, Jupyter

- Developed a Proof-of-Work blockchain simulator to evaluate throughput, latency, and energy consumption across varying transaction loads and mining difficulty levels.
- Trained ML models to predict blockchain performance metrics with up to 90% accuracy, enabling data-driven analysis of network behavior.

Multimodal Generative AI Copilot | Generative AI, Multimodal | Python, LLMs, OCR, RAG

- Built an AI copilot capable of processing PDFs, images, audio, and text to generate summaries and answer context-aware questions across multiple document types.
- Implemented OCR and RAG pipelines to improve factual grounding and reduce hallucinations by retrieving relevant context from user-uploaded documents.

ACHIEVEMENTS

Samved Hackathon 2026 — Third Prize, All India | Top 3 of 500+ teams | INR 50,000 Cash Award

AI-Driven Smart Water Distribution System | Flutter, Python, MQTT, ESP32, Machine Learning

- Developed an IoT-based water distribution system using ESP32, Arduino Mega, flow meters, and automated valves to monitor and control zone-wise water supply in real time via MQTT cloud communication.
- Built a Flutter dashboard and Python ML pipeline to forecast zone-wise demand, reducing supply imbalance by **20%** and improving allocation accuracy to **90%** across distribution zones.